

09 — PTI 50: the Gate Blocks (expected 400 error, no results)

Proves · The PTI gate blocking: the app refuses to calculate, and the message names the missing kW.

Mechanics this scenario turns on

- PTI is the shaft motor: the battery pushing power through the main engine's shaft. A gate checks that the installed PTI capacity can carry the propulsion band the battery is asked to shave; if not, the whole calculation is refused with a 400.
- Validation runs **before** Level 1. A plant that cannot carry its load is rejected there, with a different message and a different code path from the Level 1 rejections.

Panels described below · The rejection arithmetic · Expected screen · Why 400 and not 500

Anything not described here behaves exactly as in [01-excel-baseline](#) — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 08.

Test 01 + **PTI = 50**. This scenario is SUPPOSED to fail — the red banner is the pass criterion.

The rejection arithmetic

```
PTI capacity = 2 × 50 = 100 kW
Required      = battery propulsion band = 200.6 kW
100 < 200.6 → EVERY combination is rejected by the battery-PTI gate
              → zero valid combinations → NoValidCombinationException → HTTP 400
```

Expected screen

- Red "Validation Failed" banner with the QA-C-1 message containing the REAL numbers: "...the battery needs 200.6 kW of PTI capacity to shave propulsion peaks in Transit mode, but only 100 kW is available. Increase the PTI capacity per main engine (currently 50 kW), reduce the battery power, or clear the PTI field..."
- **Empty results panel** ("Results will appear here") — no stale numbers left behind.

Why 400 and not 500

An infeasible configuration is a USER-input problem, answered in the validation shape so the client renders the reason. Before fix QA-C-1 (D6) this path returned a bare 500.

Takeaway: the error is actionable — exact deficit, exact cause, three concrete ways out. Compare with test 17, which fails EARLIER (input validation) with a different message.

